Uropharmacology
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Disclosure: Financial interests (DSMB) for CG Oncology, Protara Therapeutics

Learning Objectives

1. Describe the pharmacologic management of benign prostatic hyperplasia (BPH), including indications, medications, mechanisms of action, and side effects/drug interactions

2. Describe the pharmacologic management of erectile dysfunction (ED), including indications, medications, mechanisms of action, and side effects/drug interactions

3. Describe the pharmacologic management of overactive bladder (OAB) and lower urinary tract symptoms (LUTS), including indications, medications, mechanisms of action, and side effects/drug interactions

4. Describe the pharmacologic management of hypogonadism, including indications, medications, mechanisms of action, and side effects/drug interactions

5. Describe hormonal-based therapies for prostate cancer, including indications, medications, mechanisms of action, and side effects/drug interactions

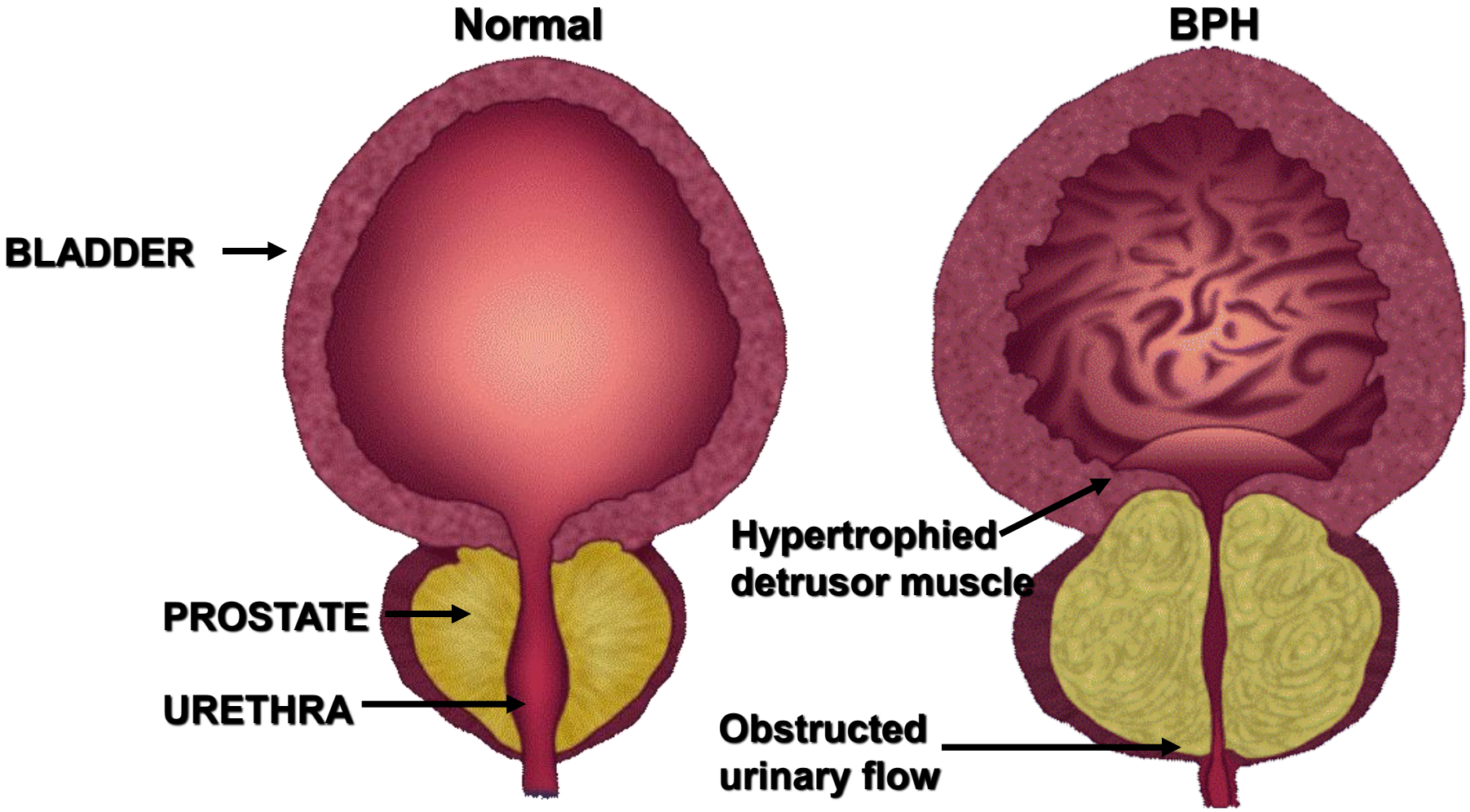
Uropharmacology Outline

- Benign Prostatic Hyperplasia (BPH)
- Erectile Dysfunction (ED)
- Voiding Dysfunction/Overactive Bladder (OAB)
- Hypogonadism
- Androgen deprivation for prostate cancer

Benign Prostatic Hyperplasia(BPH)

- Occurs in nearly 80% of males by age 80
- 1.7 million office visits/year
- 1 out of 4 males (50-80) will undergo treatment for symptoms
  - Irritative (urgency, frequency, nocturia)
  - Obstructive (poor stream, straining, incomplete bladder emptying, hesitancy)

Anatomy of BPH



Kirby RS et al. *Benign prostatic hyperplasia*. Health Press, 1995.

Benign Prostatic Hyperplasia(BPH)

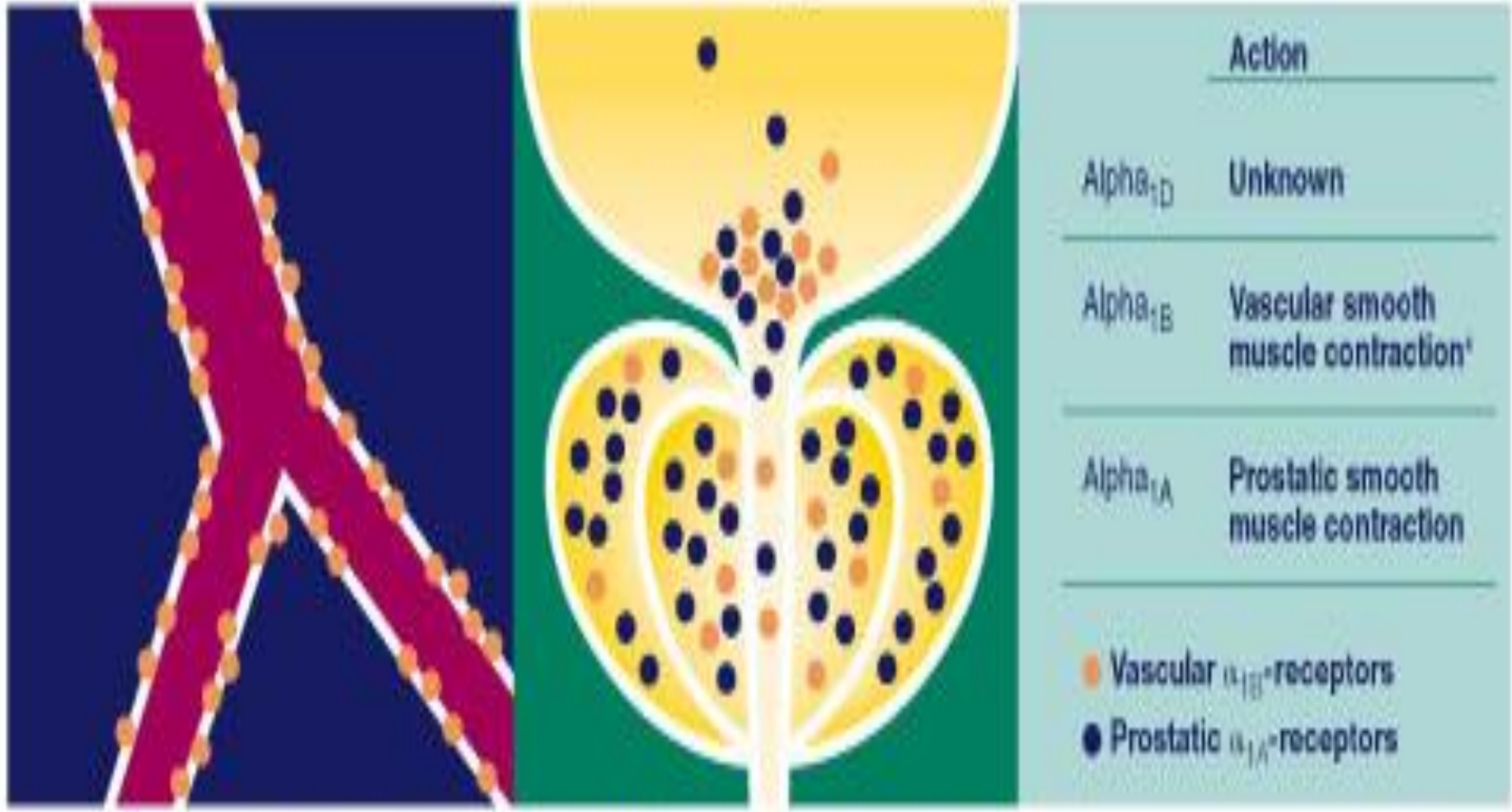
- Severe sequelae of BPH
  - UTI/urosepsis
  - Urinary retention
  - Hydronephrosis/renal failure
  - Bladder calculi
  - Hematuria/prostatic bleeding

Estimates of $$ spent in US: > $4 billion…***!!***

Drug Treatments for BPH

- **<u>Alpha antagonists</u>** –
  - Terazosin (Hytrin™), tamsulosin (Flomax ™), doxazosin (Cardura ™), alfuzosin (Uroxatrol ™), silodosin (Rapaflo ™)
  - choose based on BP and price
- **<u>5-alpha reductase</u>** inhibitors
  - Finasteride (Proscar ™) /Dutasteride (Avodart ™)-
  - may take 6-12 months for maximum effect
- **Combination Therapy**
  - Jalyn ™ (Tamsulosin and dutasteride)

- At least three discrete alpha$_1$-adrenoceptor subtypes have been identified.



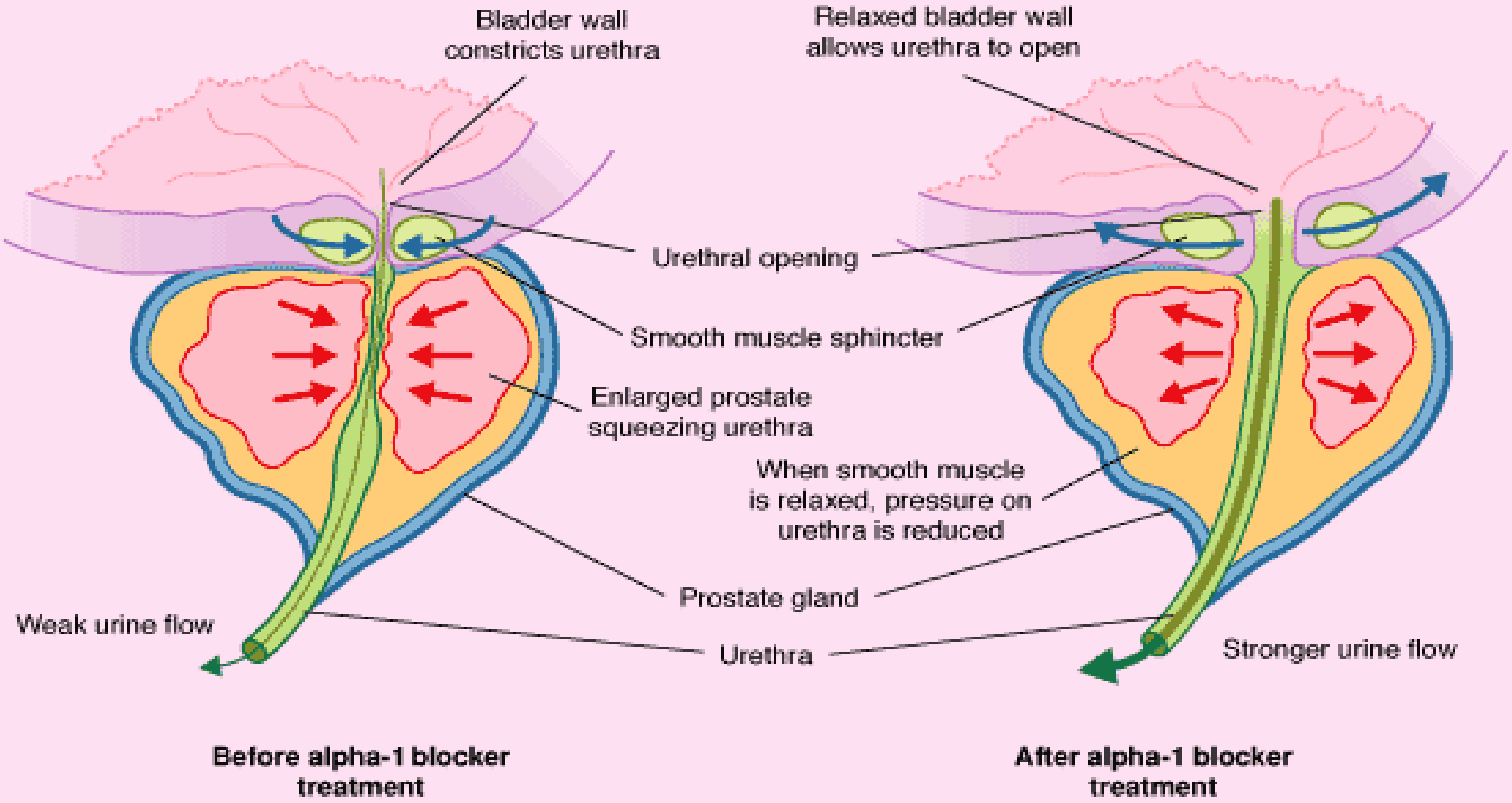
- Approximately 70% of the alpha$_1$-receptors in the human prostate are of the alpha$_{1A}$ subtype.

Clinically Useful Alpha Blockers

- Terazosin (Hytrin™)
- Doxazosin (Cardura™)

} Require dose titration

- Tamsulosin (Flomax™)
  - selective α1A
- Alfuzosin (Uroxatral™)
  - Selective α1A
- Silodosin (Rapaflo™)
  - Selective α1A


Bladder wall constricts urethra
Relaxed bladder wall allows urethra to open
Urethral opening
Smooth muscle sphincter
Enlarged prostate squeezing urethra
When smooth muscle is relaxed, pressure on urethra is reduced
Prostate gland
Urethra
Weak urine flow
Stronger urine flow
Before alpha-1 blocker treatment
After alpha-1 blocker treatment

Effects of alpha$_1$-blockers

- Typical onset of action < 1 month
- Reductions in symptom scores
  - additional effect of alpha$_1$-blockers over placebo: 15–20%
- Improvement in Urinary Flow Rate
  - additional effect of alpha$_1$-blockers over placebo: 10–15%
- Discontinuations due to adverse events
  - additional effect of alpha$_1$-blockers over placebo: 4–10%

Djavan & Marberger (1999)

First Dose Phenomenon

- Sudden and profound postural hypotension
- Typically occurs with first time taking dose, or first time after a lengthy interval off med
- Noted with first generation alpha blockers – fairly common
- Rarer but still reported with newer agents (tamsulosin)
- Can cause syncope, fall, etc
- Should counsel pts (especially elderly patients)

Alpha Blocker Side Effects

- Postural hypotension 5%
  - ➔ Dizziness 5-15%
- Nasal congestion 5%
- Headache 5-10%
- Asthenia 5-10%
- Abnormal ejaculation
  - Tamsulosin 3-10%
- Erectile dysfunction (may cause or improve ED!)
- Avoid with planned cataract surgery (Floppy Iris Syndrome)

1) Kirby R and Lepor H. In: *Campbell's Urology*, 9th edition. Philadelphia, Pa: Saunders; 2007:2766-2802. 2) Milani S and Djavan B. BJU Int 2005:95(suppl 4);29-36. 3) Patel AK and Chapple CR. BMJ. 2006 Sep 9;333(7567):535-9.

Commonly Used Alpha Blockers

Name	Side effects/notes
2nd Generation	
Terazosin (Hytrin™)	First dose syncope; dizziness; tachycardia; headache; asthenia; rhinitis
Doxazosin (Cardura™)	Same as above
3rd Generation	
Alfuzosin (Uroxatral™)	Dizziness; headache; minimal cardiovascular effect; ejaculatory dysfunction (less than tamsulosin)
Tamsulosin (Flomax™)	Ejaculatory dysfunction; rhinitis
Silodosin (Rapaflo™)	Retrograde ejaculation

Titrate dose weekly up to desired dose

BPH Progression

- In many men, BPH is a progressive condition:
  - Increase in prostate volume
  - Worsening of symptoms
  - Deterioration in urinary flow rate
  - Increased risk of acute urinary retention (AUR)
  - Increased risk of needing surgery for BPH

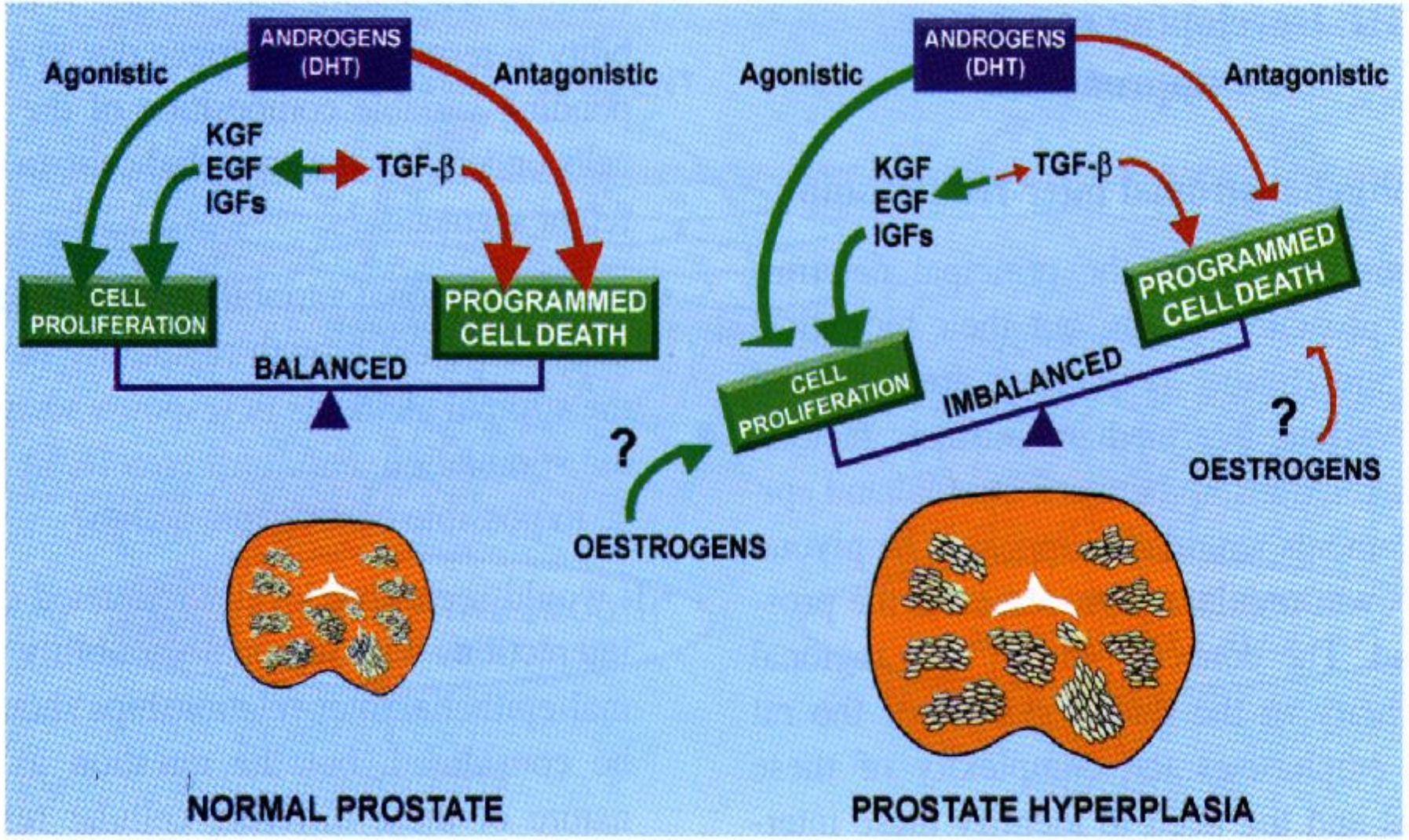
- *Prostate enlargement in BPH is driven primarily by androgens*

Dihydrotestosterone Is the Primary Androgen Responsible for Prostate Growth

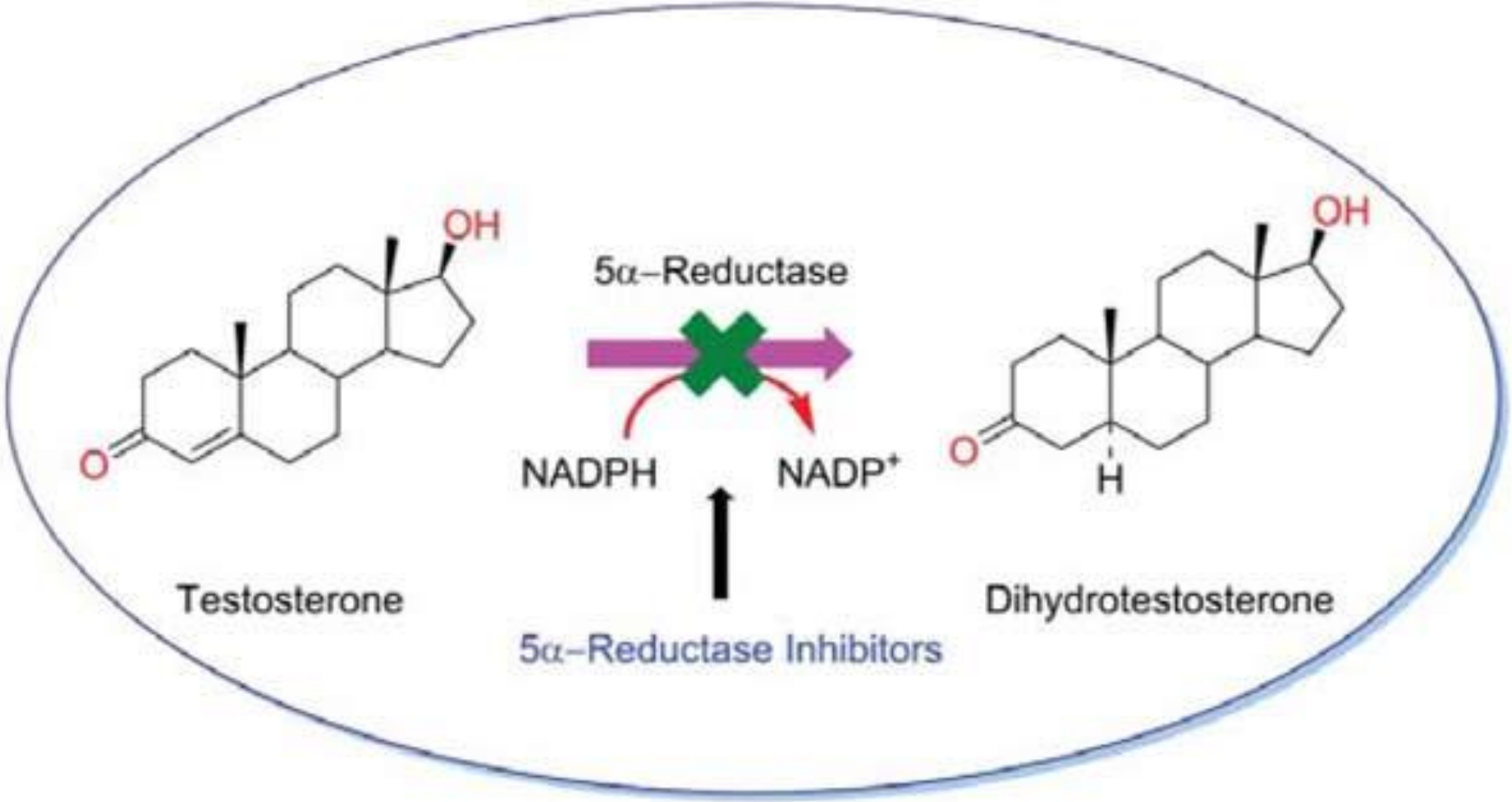
- DHT is important in fetal development and continues to promote prostate growth through puberty
- In adult males, DHT maintains the balance between cell proliferation and cell death

Bartsch G et al. *Eur Urol*. 2000;37:367-380.
Griffiths K. Chapter 3 in *Textbook of Benign Prostatic Hyperplasia*. Isis Medical Media Ltd, 1996.

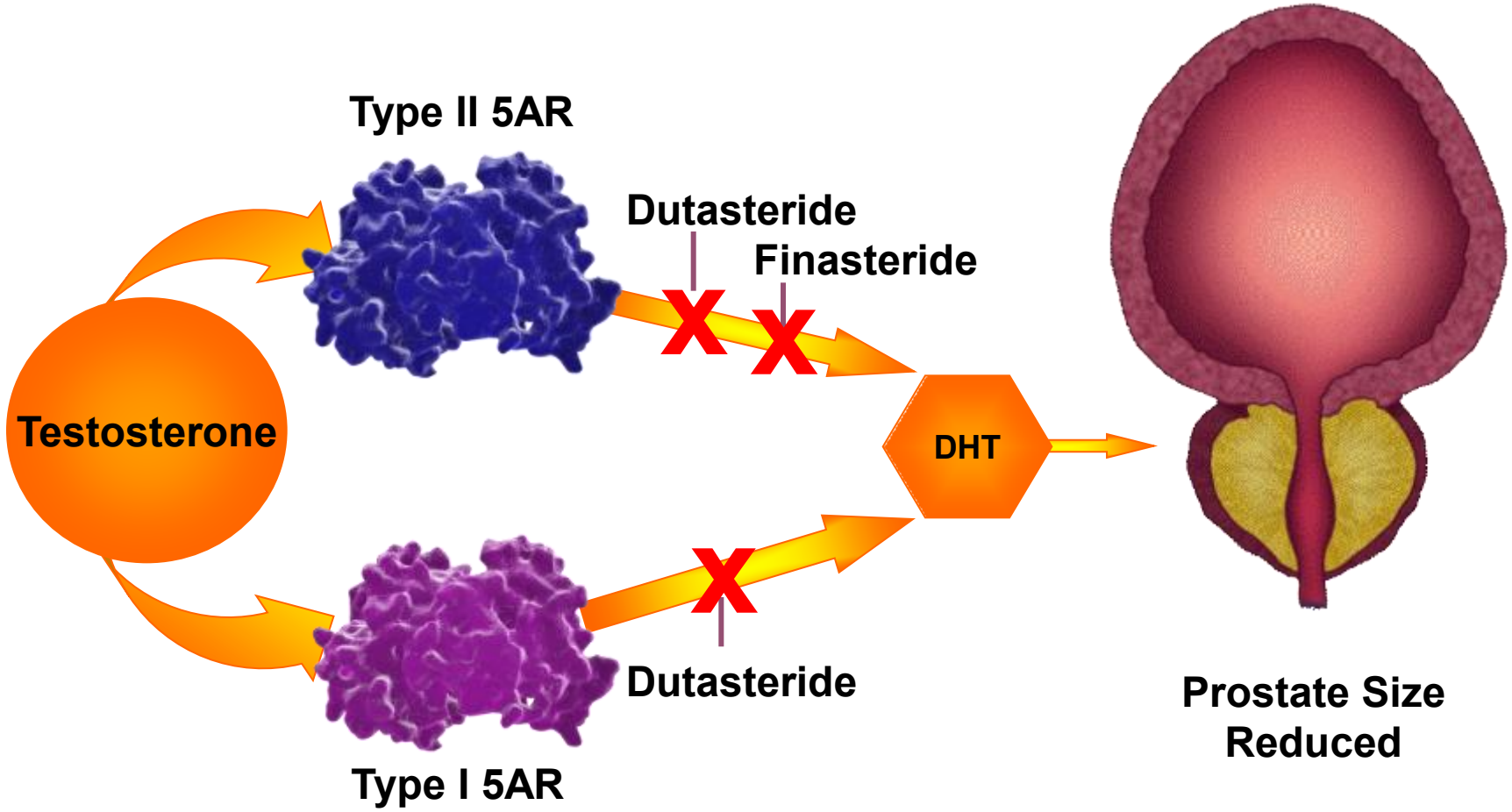
Impact of Androgens on Prostate



Testosterone is Converted to DHT via Type I and II 5α-Reductase



5-α Reductase Inhibitors (5ARI's)



5-AR = 5-alpha reductase; DHT = dihydrotestosterone

1) Steers WD. *Urology*. 2001;58(suppl 6A):17–24. 2) Bartsch G et al. *Eur Urol*. 2000;37:367–380.

5α-Reductase Isoenzymes

	Type 1	Type 2
Chromosome location	5	2
Tissue distribution	Predominately liver and skin	Predominately genital tissue
Found in prostate?	Yes	Yes
Inhibition by finasteride	No	Yes
Inhibition by dutasteride	Yes	Yes

Bartsch G et al. *Eur Urol.* 2000;37:367-380.

5 Alpha Reductase Inhibitors

Name	Half-life	Mechanism
Finasteride (Proscar™)	6-8 hrs	Inhibits type II 5-AR
Dutasteride (Avodart™)	3-5 weeks	Inhibits type I and II 5-AR

5 Alpha Reductase Inhibitors

Side Effects

• Decreased libido	3-8%
• Ejaculatory disorder	1-5%
• Erectile dysfunction	5-10%
• Breast tenderness	1%
• Gynecomastia	1%

Side effects are mild and decrease with duration of therapy

Comparison of 5αRIs and alpha$_1$-blockers

5αRIs	Alpha$_1$-blockers
• Symptom improvement from ~6 months	• Symptom improvement from < 1 month
• ↑ flow from 1 month	• ↑ flow from < 1 month
• Effects on symptoms and flow demonstrated to 10 yr	• Effect on symptoms and flow > 12 months not well established
• Reduce prostate volume	• Do not reduce prostate volume
• Reduce the risk of AUR	• Do not prevent AUR and surgery (Delay??)
• Reduce the risk of surgery	
• Decrease PSA (<1/2)	

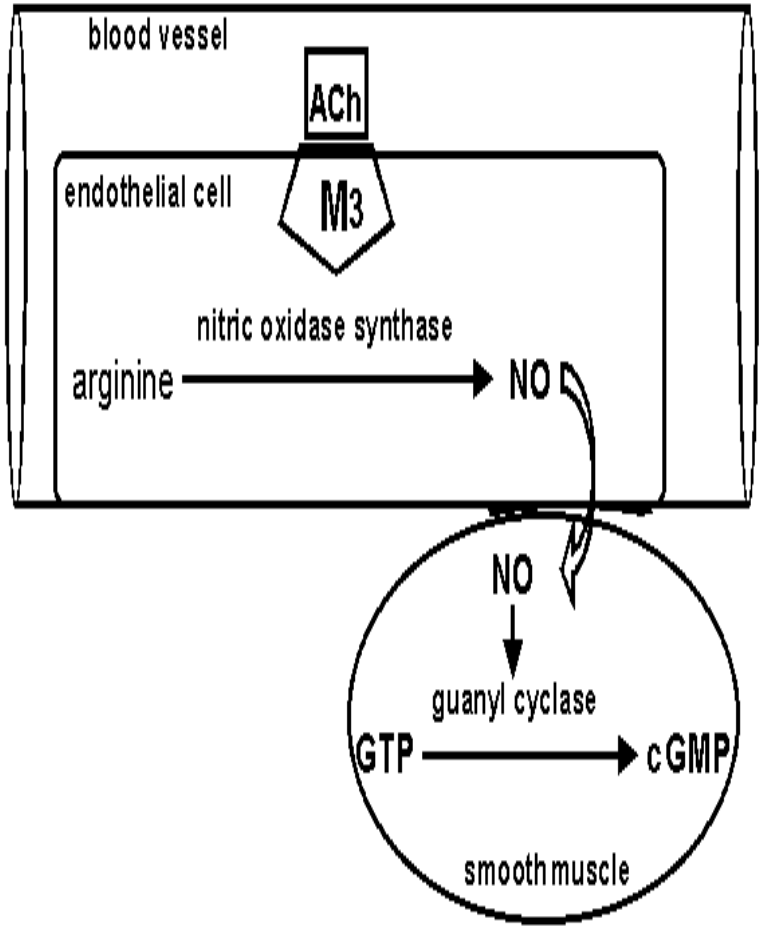
Clinical Pharmacology of Erectile Dysfunction

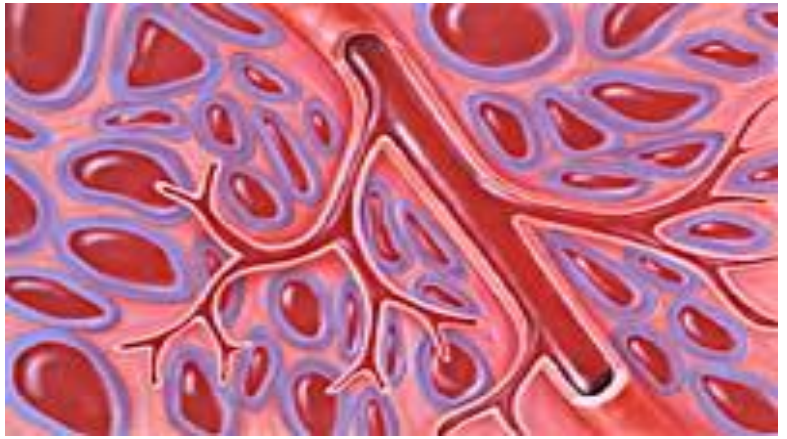
Mechanics of Erection

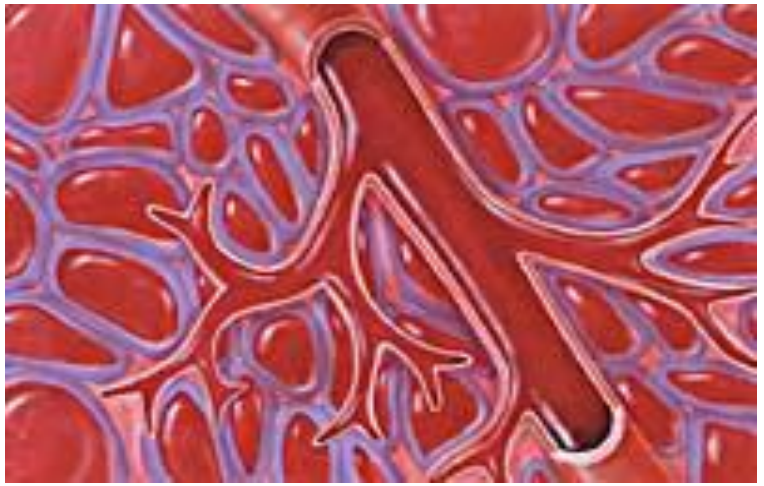
- Upon sexual stimulation, the terminals of parasympathetic nerves release nitric oxide.
- The nitric oxide diffuses into the smooth muscle cells and bind with the enzyme guanylate cyclase.

- **Guanylate cyclase** converts guanosine triphosphate (GTP) into cyclic guanosine monophosphate (cGMP).

NITRIC OXIDE MEDIATED VASODILATION







cyclic GMP → smooth muscle relaxation → vasodilation

Phosphodiesterase 5 (PDE_5) breaks down cGMP

First-Line Therapy: Lifestyle Modification

- Stop smoking[1,2]
- Limit or avoid alcohol[1]
- Follow healthy diet[2]
- Exercise regularly[3,4]

1. Lue TF et al. In: Lue TF et al, eds. *Sexual Medicine: Sexual Dysfunctions in Men and Women*. Paris, France: Health Publications; 2004:605-627. 2. Feldman HA et al. *Prev Med*. 2000;30:328-338. 3. Derby CA et al. *Urology*. 2000;56:302-306. 4. Lewis RW et al. In: Lue TF et al, eds. *Sexual Medicine: Sexual Dysfunctions in Men and Women*. Paris, France: Health Publications; 2004:37-72.

First-Line Therapy: Drug Therapy Modifications

- Modify drug regimens associated with ED[1]
  - Antihypertensives/diuretics[2]
  - Selective serotonin-reuptake inhibitors (SSRIs)[3]
  - Hormonal agents (eg, antiandrogens)[4,5]
  - H_2-receptor antagonists[6]

1. Lue TF. *N Engl J Med*. 2000;342:1802-1813. 2. Grimm RH Jr et al. *Hypertension*. 1997;29:8-14. 3. Rosen RC et al. *J Clin Psychopharmacol*. 1999;19:67-85. 4. Jackson G et al. *Int J Clin Pract*. 1999;53:445-451. 5. Ralph D et al. *BMJ*. 2000;321:499-503. 6. Lundberg PO, Biriell C. *Int J Impot Res*. 1993;5:237-239.

First-Line Therapy: Testosterone Replacement

- Most patients **do not** have testosterone deficiency as cause of ED

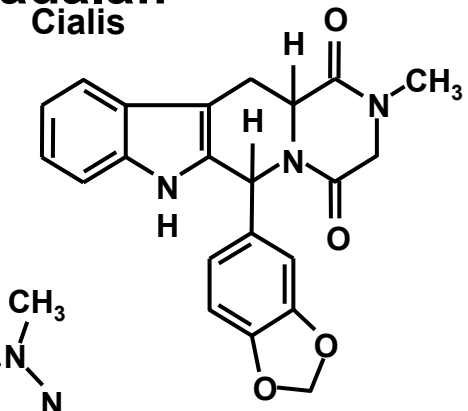
- Use **only** if this is documented on multiple AM blood draws

- Formulations:
  - Transdermal testosterone[1,2]
    - Topical gel and patches
  - Intramuscular (IM) injection (eg, testosterone cypionate)[2]
  - Subcutaneous implant[3]
  - Oral testosterone now available in United States (previously banned due to hepatotoxicity[1,2])

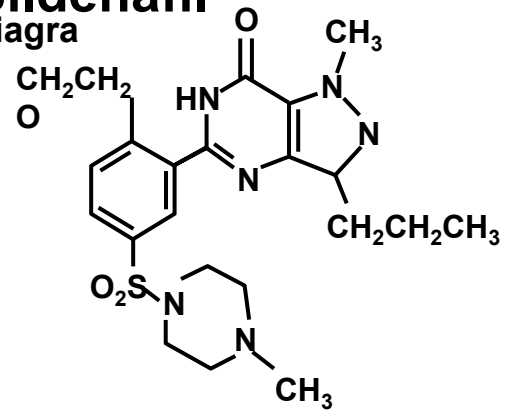
1. AACE Male Sexual Dysfunction Task Force. *Endocr Pract*. 2003;9:77-95. 2. Lue TF. *N Engl J Med*. 2000;342:1802-1813. 3. Testopel® (testosterone) pellets prescribing information. Bartor Pharmacal Co. Inc: Rye, NY; 1992.

First-Line Therapy: Oral PDE5 Inhibitors

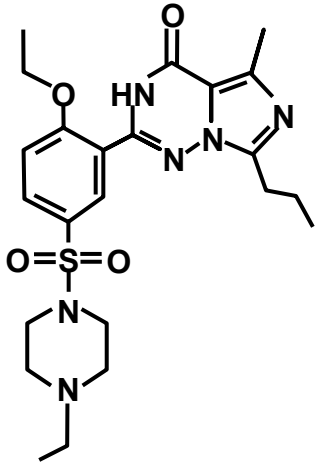
Tadalafi[2]
Cialis



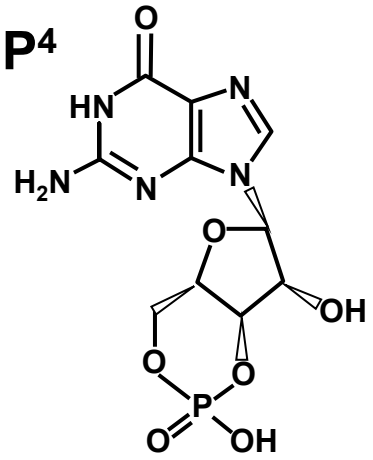
Sildenafil[1]
Viagra



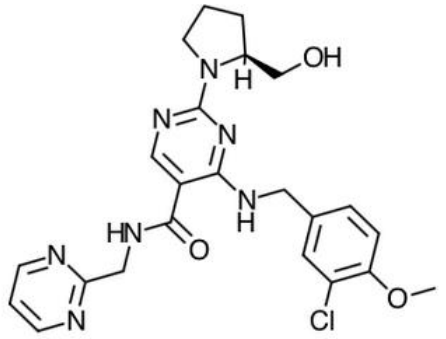
Vardenafil[3]
Levitra



cGMP[4]

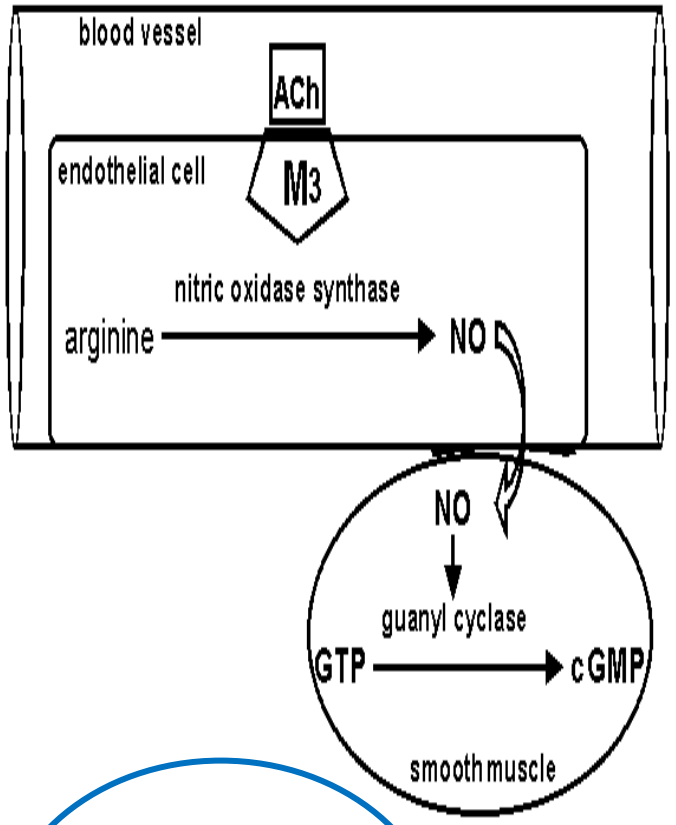


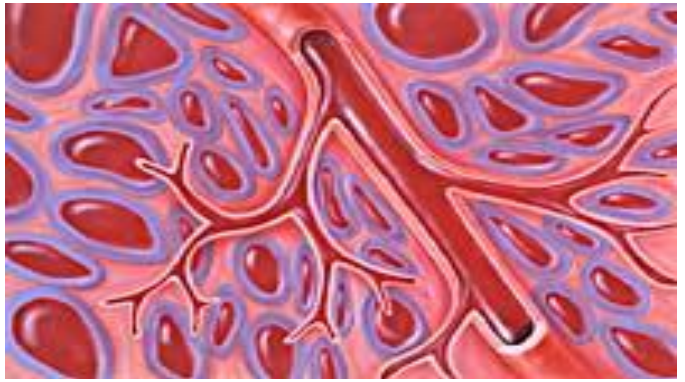
Avanafil
Stendra

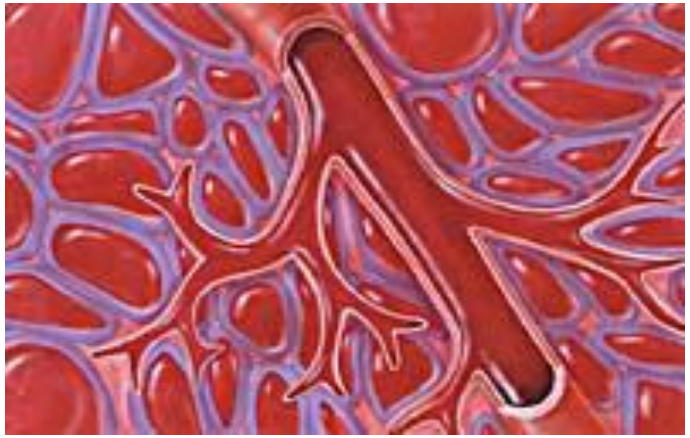


1. Viagra® (sildenafil) prescribing information. Pfizer Inc: New York, NY; 2005. 2. Cialis® (tadalafil) prescribing information. Lilly ICOS LLC: Indianapolis, Ind and Bothell, Wash; 2005. 3. Levitra® (vardenafil) prescribing information. Bayer Pharmaceuticals Corp: West Haven, Conn; 2005. 4. Corbin JD, Francis SH. *Int J Clin Pract*. 2002;56:453-459.

Nitric Oxide-mediated Vasodilation







cyclic GMP → smooth muscle relaxation → vasodilation

Phosphodiesterase 5 (PDE_5) breaks down cGMP

PDE5 Inhibitors

Name	Time to Onset	Serum Half Life	Affected by food?
Sildenafil (Viagra™)	60 min	4 hours	Yes (delays absorption/ onset)
Vardenafil (Levitra™)	60 min	4 hours	Yes
Tadalafil (Cialis™)	120 min	17.5 hours	No
Avanafil (Stendra™)	60 min	5 hours	No

Adverse Events: PDE5 Inhibitors

Adverse Event	Reported by ≥2% of Patients (%)		
	Sildenafil	Tadalafil	Vardenafil
Headache	16	15	15
Flushing	10	3	11
Rhinitis	—	—	9
Nasal congestion	4	3	—
Dyspepsia	7	10	4
Abnormal vision	3	—	—
Sinusitis	—	NR	3
Flu syndrome	>2	NR	3
Diarrhea	3	—	—
Myalgia	—	3	—
Back pain	>2	6	2

PDE5 Inhibitor Contraindications

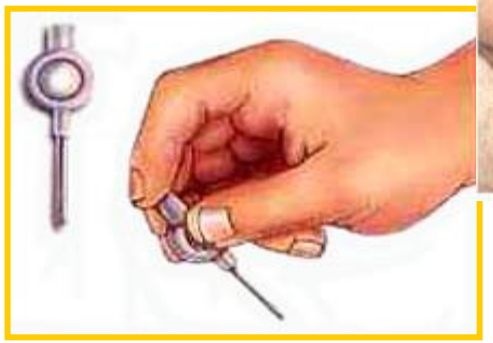
- PDE5i contraindicated with nitrates
  - Can potentiate vasodilation and hypotension associated with nitric oxide
  - Several early deaths in elderly men with concomitant heart disease on nitrates
- Caution with use in patients with vascular disease, coronary disease or vascular risk factors
  - Cardiac eval before starting PDE5i

High-Risk Patient: Princeton Consensus Panel Recommendations

- Unstable or refractory angina
- Uncontrolled HTN
- Congestive heart failure (CHF; NYHA class III/IV)
- Recent MI (<2 wk)
- High-risk arrhythmias
- Obstructive hypertrophic cardiomyopathy
- Moderate/severe valvular disease

Kostis JB et al. *Am J Cardiol*. 2005;96:313-321.

Second-Line Therapy: MUSE Transurethral System





- Lack of response to oral therapy[1]
- Contraindications or adverse reactions to specific oral drugs[1]
- **Intraurethral Alprostadil (PGE-1)**





- **Penile Injection therapy**
  - **Alprostadil (PGE-1)**
  - **Papaverine**
  - **Phentolamine**

1. Lue TF et al. In: Lue TF et al, eds. *Sexual Medicine: Sexual Dysfunctions in Men and Women*. Paris, France: Health Publications; 2004:605-627. 2. MUSE® (alprostadil) urethral suppository prescribing information. Vivus Inc: Mountain View, Calif; 1998. 3. Benevides MD, Carson CC. *J Urol.* 2000;163:785-787. 4. Engel JD, McVary KT. *Urology*. 1998;51:687-692

Transurethral/Intracavernosal Therapy

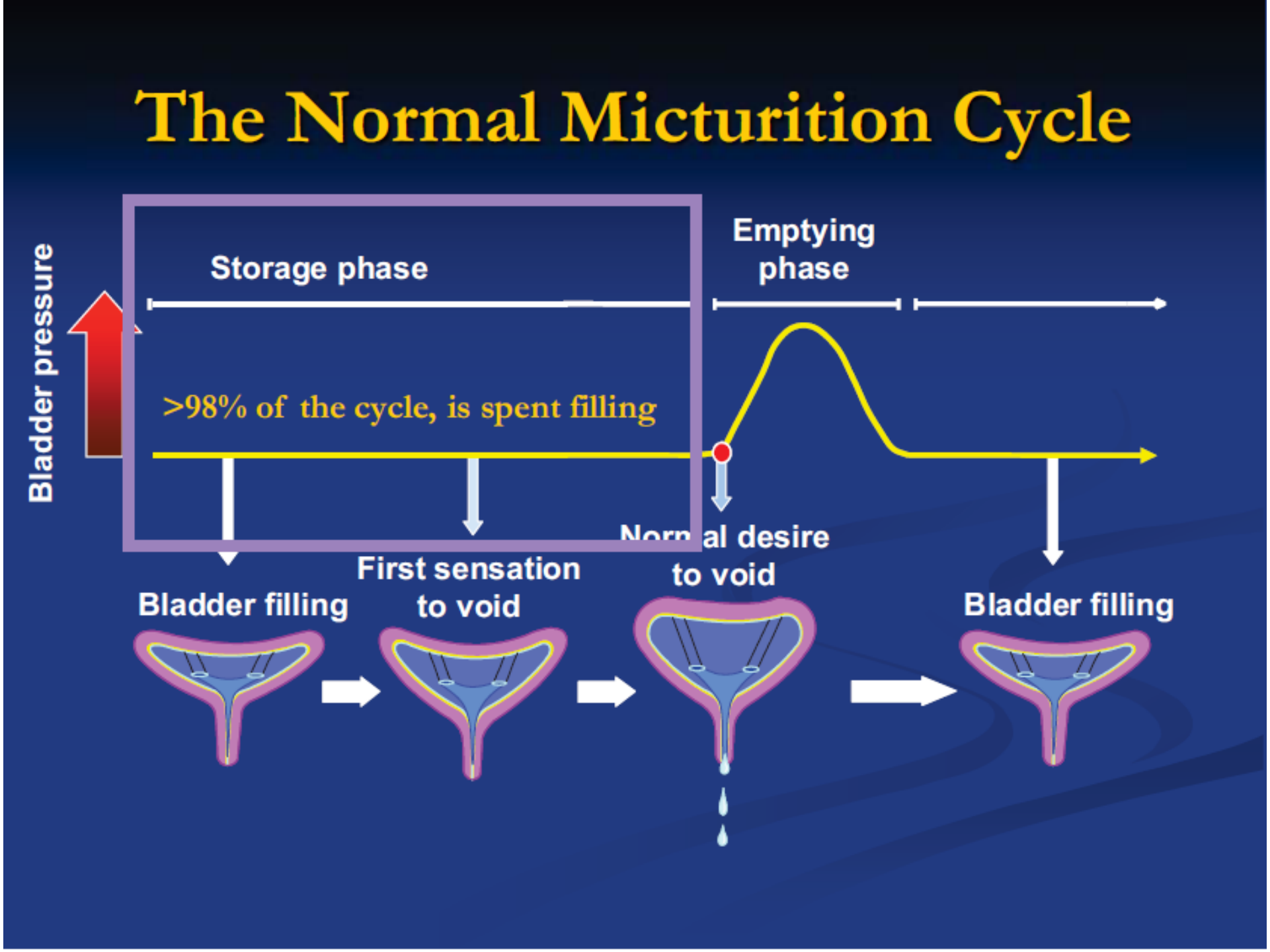
Name	Mechanism of Action	Side effects/notes
Alprostadil TU (MUSE™)	Stimulates adenyl cyclase to ⇑cAMP	Painful erection; urethral pain or bleeding; priapism (rare)
Alprostadil IC (Cavarject™, Edex™)	Same as above	Pain; fibrosis; hematoma; priapism (rare)
Papaverine IC	Non-select PDE5i; ⇑cAMP & cGMP	Priapism; corporal fibrosis
Phentolamine IC	Alpha blocker; Smooth muscle dilation; ⇓ sympathetic tone	Hypotension; reflex tachycardia

Pharmacology of Voiding Dysfunction / Overactive Bladder (OAB)

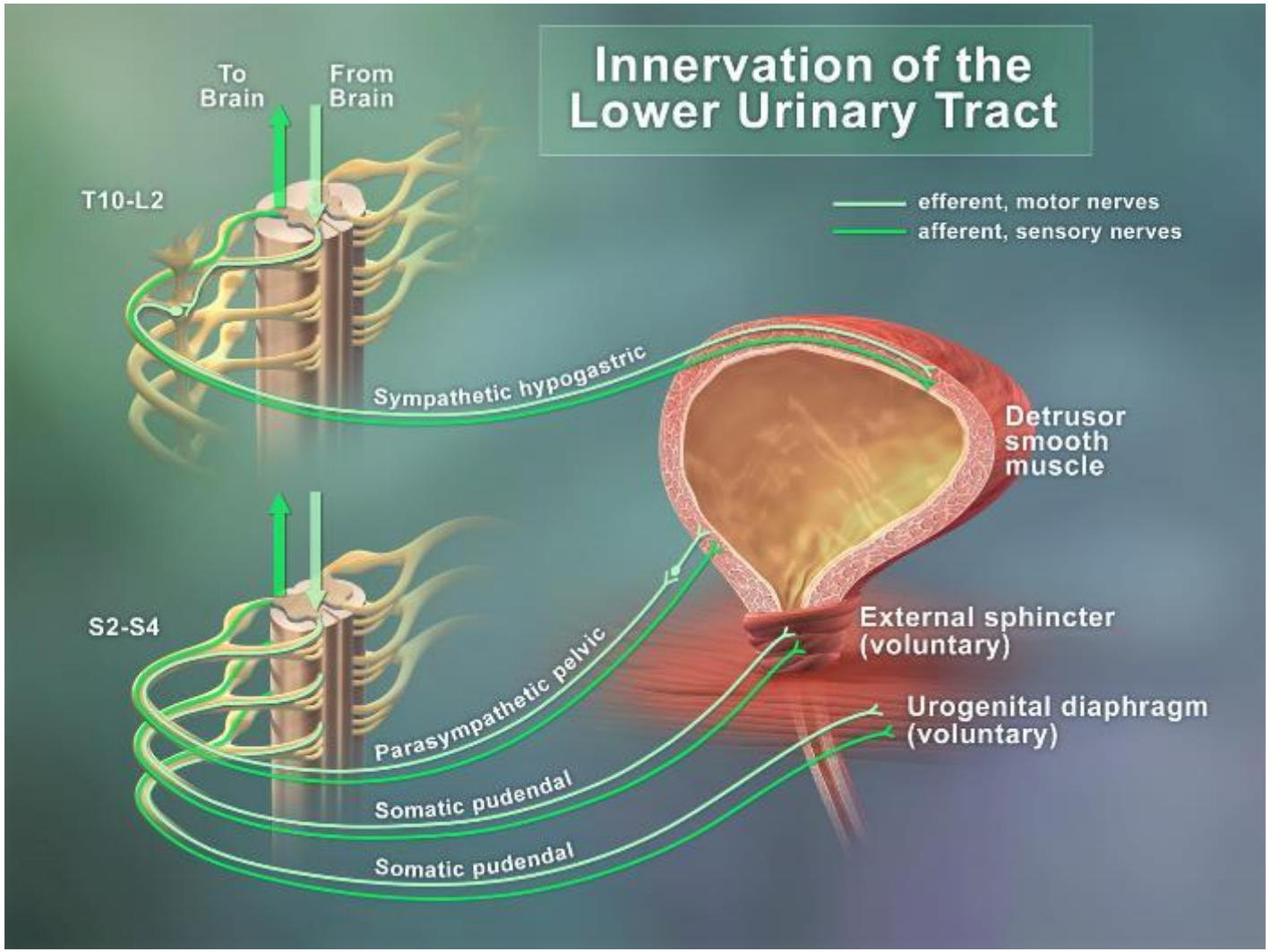
Anatomy & Neurophysiology of the Lower Urinary Tract

- Bladder (detrusor)
  - Stores urine at low pressure
  - Compresses urine for voiding
- Urethra
  - Conveys urine from bladder to outside world
- Sphincter(s) internal & external
  - Controls urine flow & maintain continence between voidings

The Normal Micturition Cycle



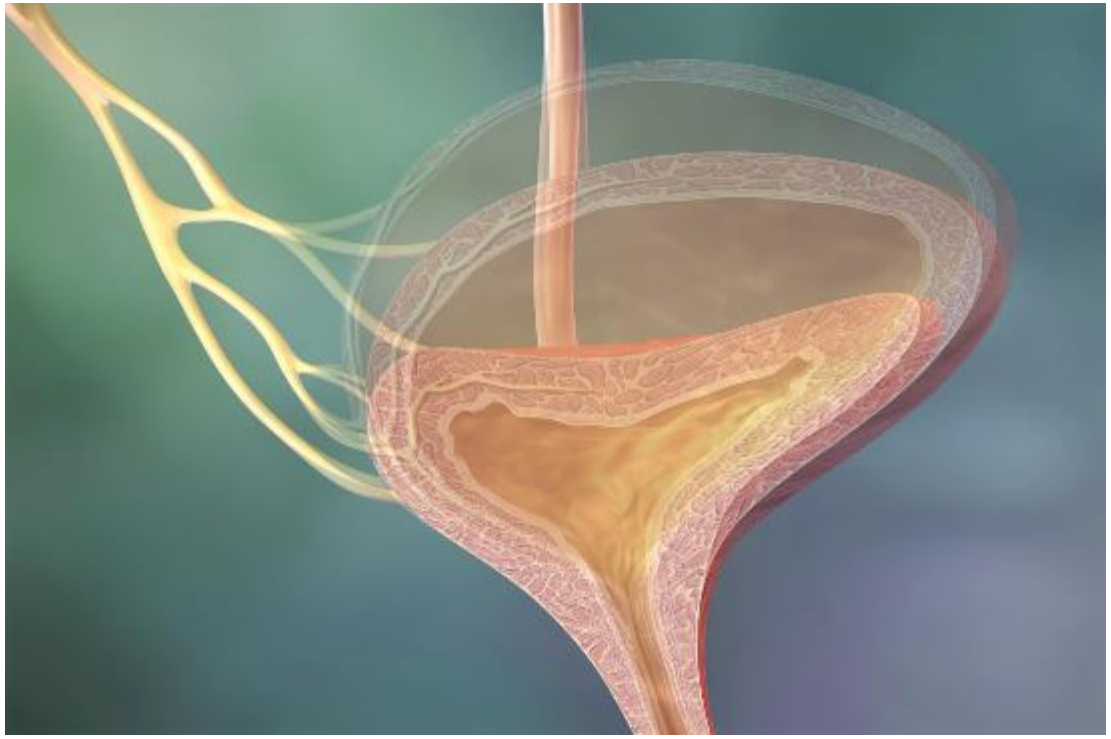
Neural Control of the Lower Urinary Tract



- **Micturition controlled by central and peripheral nervous systems**
- **Acetylcholine (Ach) primary neurotransmitter for detrusor contraction**

Andersson K-E, Hedlund P. *Urology*. 2002;60(suppl 5A):13-21.

Detrusor Muscle Contraction in Patients With OAB May Lead to Uncontrolled Micturition



- ACh may be released from neuronal and non-neuronal sources (eg, urothelium)
- ACh may then directly or indirectly excite afferent nerves in suburothelium and detrusor, and trigger urgency/frequency
- The muscarinic receptor family comprises five subtypes

Andersson K-E. *Lancet Neurol*. 2004;3:46-53.
Andersson K-E, Hedlund P. *Urology*. 2002;60(suppl 5A):13-21.
Andersson K-E, Yoshida M. *Eur Urol*. 2003;43:1-5.

Distribution and Function of Muscarinic Receptors Throughout the Body

	Location	Function
M_1	Cerebral cortex, hippocampus, salivary glands, eye	Memory and cognitive function, saliva and tear secretion
M_2	**Smooth muscle**, hippocampus, hindbrain, cardiac muscle, eye	Heart rate, tear secretion
M_3	**Smooth muscle**, salivary glands, eye, brain	Bladder contraction, bowel motility, saliva and tear secretion, visual accommodation
M_4	Basal forebrain striatum, salivary glands	Unknown
M_5	Substantia nigra, eye	Visual accommodation



M_3 Receptor Is the Primary Mediator of Bladder Contraction

- Membrane-bound proteins activated by acetylcholine (ACh)
- Normal bladder contraction mediated by activation of muscarinic receptors in the detrusor muscle
- M_3 shown in numerous studies to be primary mediator of bladder contraction despite M_2 being more prevalent than M_3 in the bladder

Andersson K-E, Yoshida M. *Eur Urol*. 2003;43:1-5.
Chapple CR et al. *Urology*. 2002;60(suppl 5A):82-89.
Choppin A, Eglen RM. *Br J Pharmacol*. 2001;132:835-842.

Antimuscarinic Overactive Bladder Medications

- Mechanism of action
  - Antimuscarinic action
    - Inhibits binding of acetylcholine to cholinergic receptor and suppresses bladder contractions
  - Antispasmodic
    - Allows detrusor smooth-muscle relaxation to control overactive bladder and urge urinary incontinence

Source: Yarker YE et al. *Drugs Aging*. 1995;6:243-262.

Therapeutic Uses of Muscarinic Antagonists

- Urinary Bladder
  - Motor and sensory
    - Increase bladder capacity
    - Reduce urinary frequency
    - Diminsh urinary incontinence episodes resulting from involuntary bladder contractions

Overactive Bladder Medications
Antimuscarinic Agents

- Oxybutynin (Ditropan™)
- Oybutynin SR (Ditropan XL™)
- Tolterodine (Detrol™)
- Darifenacin (Enablex™)
- Solifenacin (Vesicare™)
- Trospium (Sanctura XR™)
- Fesoterodine (Toviaz™)

No Difference in Efficacy,

Minor differences in side effect profiles

OXYBUTYNIN Sustained Release:
Tolerability and Safety

Adverse Event	Incidence
Dry mouth	61%
Headache	10%
Constipation	13%
Diarrhea	9%
Pain	7%
Dyspepsia	7%

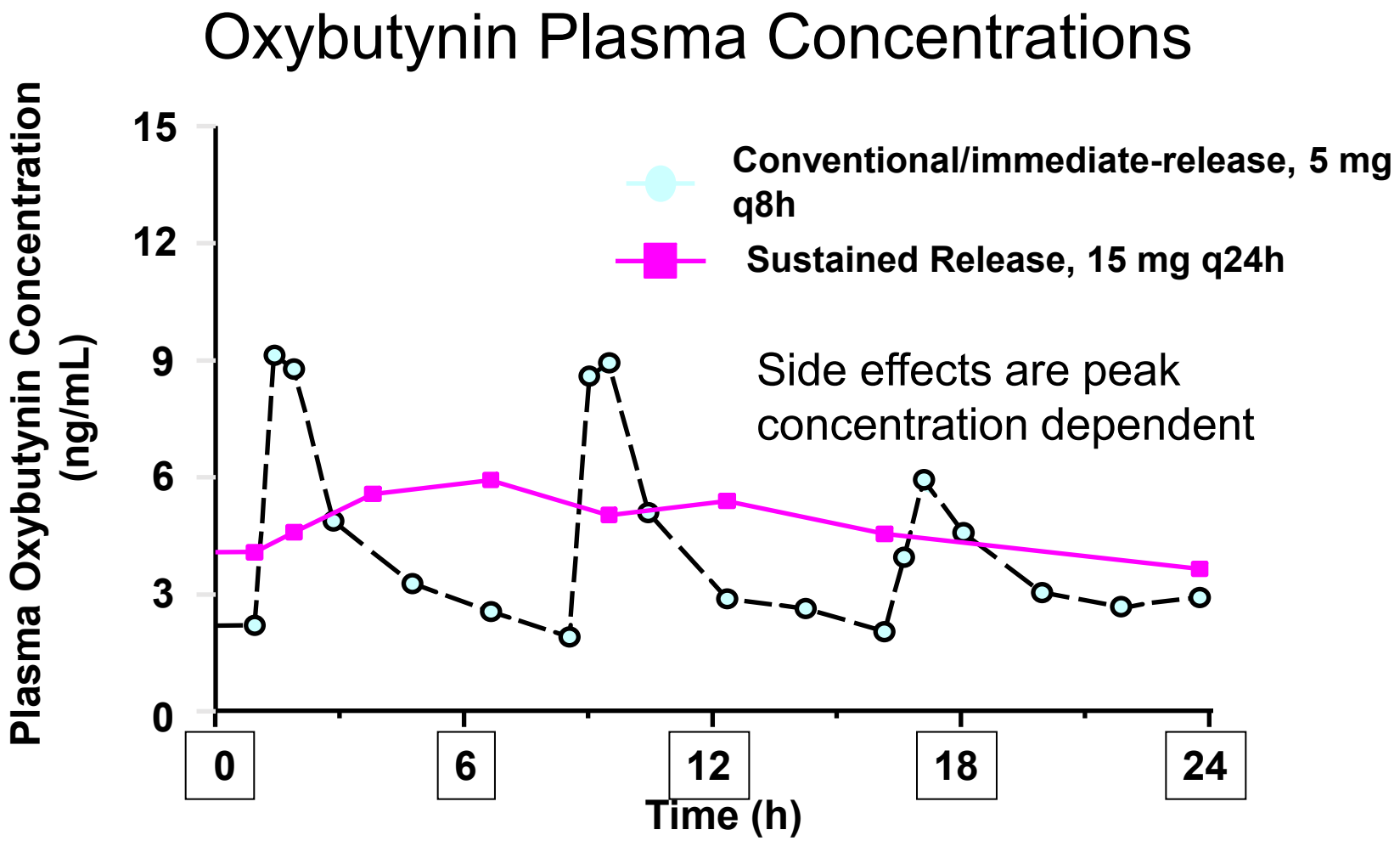
Ditropan XL® [package insert]. 2000.

Tolterodine: Tolerability

Adverse Event/Intensity	Treatment Group	
	Placebo (n=176) %	Tolterodine (n=474) %
Dry mouth	15.9%	39.5%
Headache	7.4%	11.0%
Constipation	4.5%	6.5%
Dyspepsia	1.7%	5.9%
Somnolence	1.7%	3.0%

Appell RA. *Urology.* 1997;50(suppl 6A):90-96.
Data on file. ALZA Corporation.
Detrusitol® International Monograph.

Oxybutynin SR: Comparative Pharmacokinetics



Gupta SK. *J Clin Pharmacol*. 1999;39:289-296.

Antimuscarinic therapy

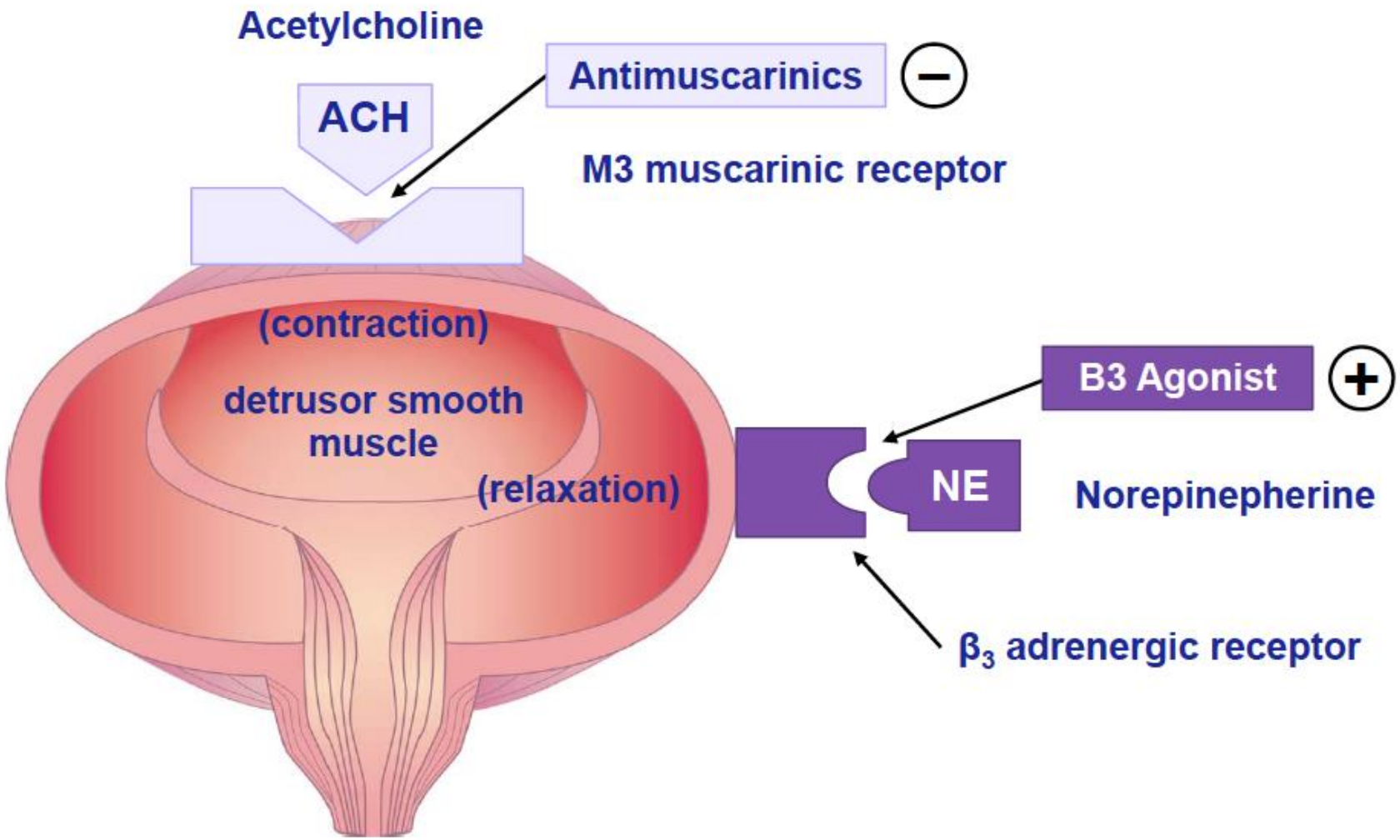
- Trospium – quaternary amine
  - Less likely to cross blood-brain barrier
  - Lower cognitive side effects

- Side effects are common reason for noncompliance/discontinuation

β3 Andrenergic Agonists

- Mirabegron (Mybetriq™) and Vibegron (Gemtesa™)
  - Novel mechanism of action
  - Significant improvement of OAB sx (urgency, frequency, urge incontinence)
  - Different side effect profile
    - No dry mouth, constipation, etc

Neurologic Innervation and Control of Bladder Musculature



Hypogonadism & Testosterone Replacement

Hypogonadism

- Serum testosterone (T) peaks in mid teens, starts decline in 30's-40's
  - 10% per decade decline in T per decade >40
- Symptoms:
  - ED, low libido, depression, fatigue, anemia
- Diagnosis both clinical and serological
  - Multiple early AM serum measurements
- Consider endocrine evaluation before T replacement
- Need to be careful in prostate cancer

T Replacement

- Oral androgens historically not used in US
  - Difficulty with achieving consistent levels
  - Hepatotoxicity
  - New oral agents: testosterone undecanoate
- Injectable agents
- Transdermal/topical agents
- Buccal preparations
- Long acting implants

Injectable Androgen

- Testosterone cypionate (Depo-Testosterone™) IM
- Testosterone enanthate (Delatestryl™) IM
- Testosterone undecanoate (Nebido™) IM

Transdermal/Topical Androgen

- Testosterone patch (Androderm™)
- Testosterone gel 1% (AndroGel™) or Testosterone gel 1% (Testim™ Fortesta™, Axiron™, Vogelxo™

Testosterone Implant

- Testosterone implant (Testopel™)
  - Implant pellets subcutaneous every 3-6 months
  - Implant in buttock with local anesthesia

Nasal and Oral Androgen

- Intranasal testosterone (Natesto™)
  - Nasal gel preparation for inhalation
  - 1 pump three times daily
  - Quickly achieve maximum serum levels (40 minutes)

  Limited long-term data

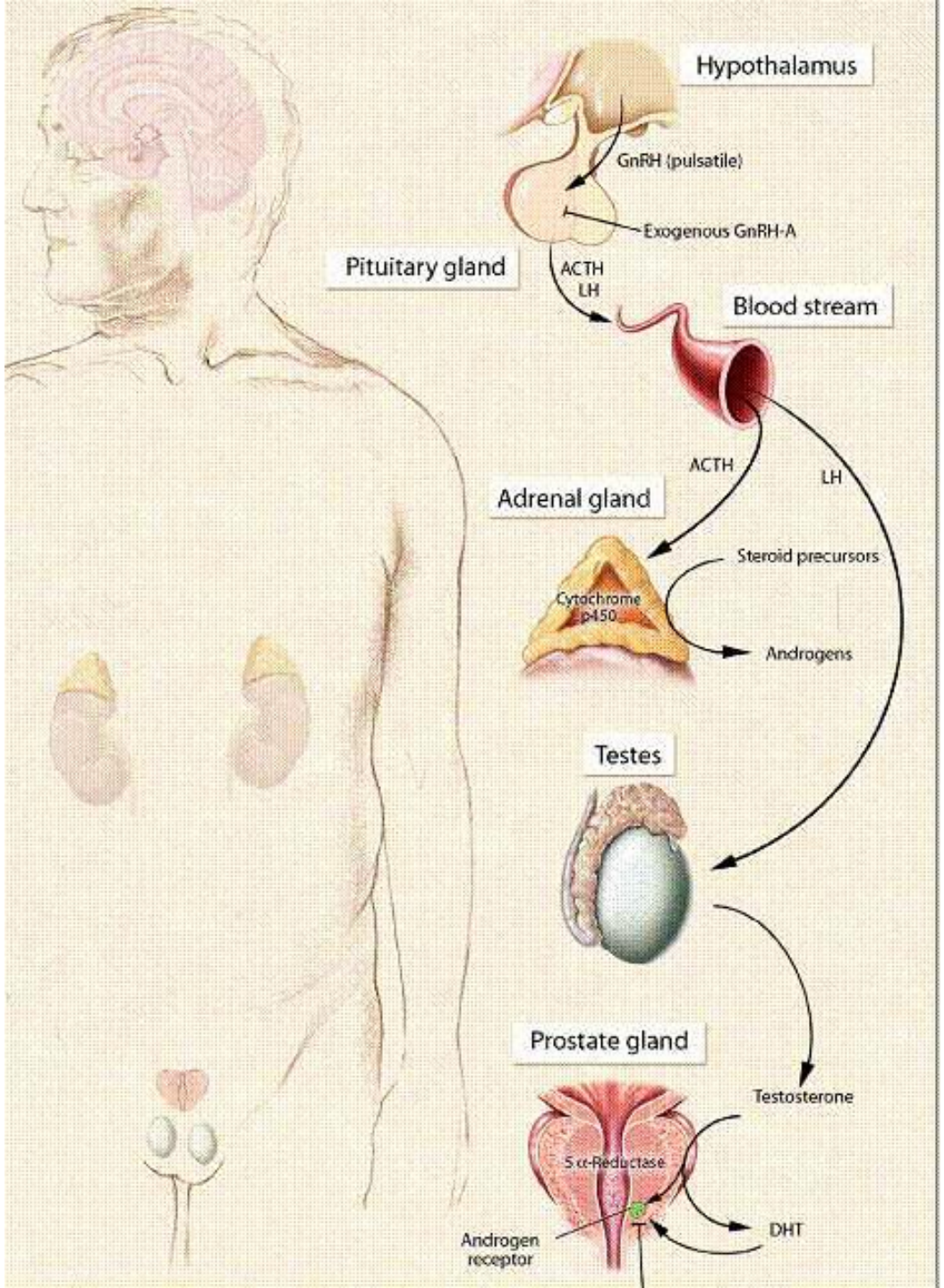
- Oral testosterone undecanoate (Jatenzo™, Kyzatrex™, Tlando™)
  - Twice daily dosing
  - Titrate based on serum levels

Androgen Deprivation

for Prostate Cancer

Androgen Deprivation

- Mainstay of treatment for advanced, metastatic and/or recurrent prostate cancer
- Androgen (primarily T) key driver of prostate growth and homeostasis
  - Castration induces cell death/apoptosis
  - PSA levels decline w/castration, correlate with prostate cancer activity
- Charles Huggins 1941 – castration in prostate cancer men (Nobel in 1966)



Androgen Axis

Hypothalamus:
- releases GnRH (pulsatile)

Pituitary:
- releases LH

Gonads:
- produces T

Constant LH (non-pulsed):
- high T, shuts down axis

Continuous Leuprolide therapy

Initial Response

GnRH Receptor stimulation

↑FSH release

↑LH release

↑ Estradiol & Progesterone (women)

↑ Testosterone (men)

Chronic Effect (2-4 weeks) *

GnRH Receptor desensitization

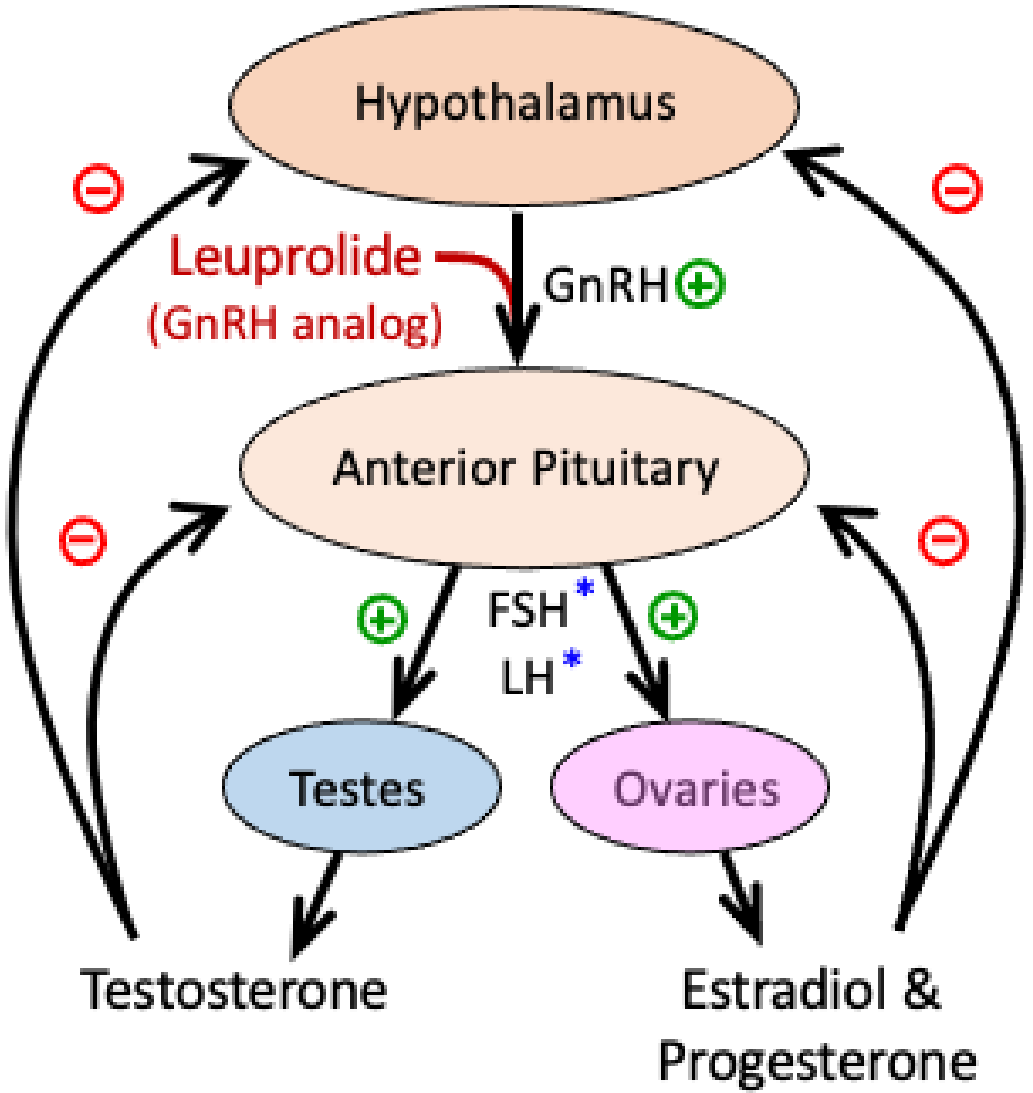
Hypogonadism

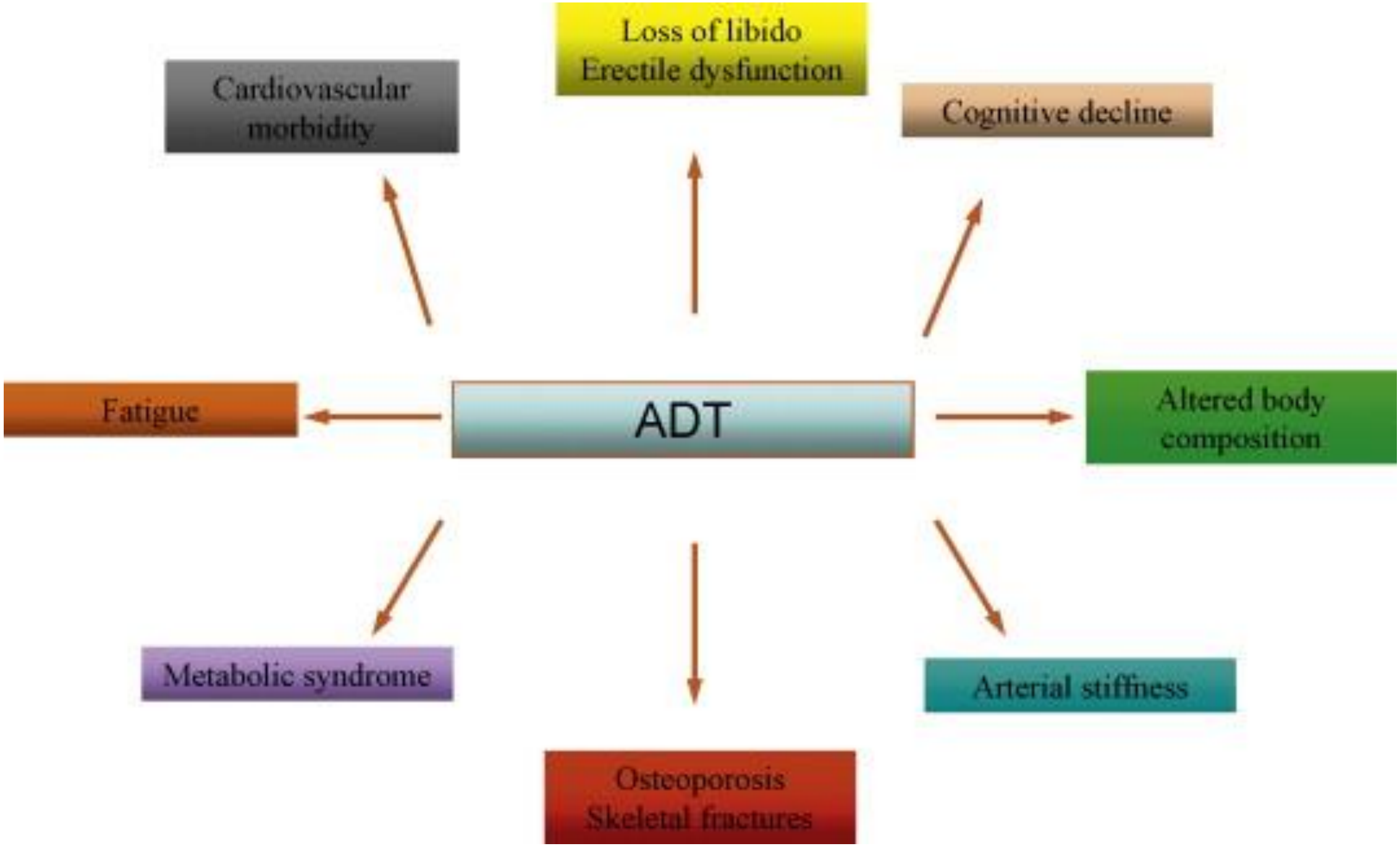
↓FSH release

↓LH release

↓Estradiol & Progesterone (women)

↓Testosterone (men)




Loss of libido
Erectile dysfunction
Cardiovascular morbidity
Cognitive decline
Fatigue
ADT
Altered body composition
Metabolic syndrome
Osteoporosis
Skeletal fractures
Arterial stiffness

LHRH Agonists/Antagonist

Name	Class	Administration	Note
Leuprolide (Lupron Depot™)	LHRH agonist	IM monthly, different intervals	Can cause initial testosterone surge
Leuprolide gel (Eligard™)	LHRH agonist	SC monthly, different intervals	Can cause initial testosterone surge; Requires refrigeration
Goserelin (Zoladex™) (Zoladex LA™)	LHRH agonist	SC monthly, different intervals	Can cause initial testosterone surge
Degarelix (Firmagon™)	LHRH antagonist	SC in 2 divided doses (loading dose) Then SC monthly	No initial testosterone surge
Relugolix (Orgovyx™)	GnRH antagonist	PO loading dose then 120 mg po daily	No initial testosterone surge

Oral Antiandrogens

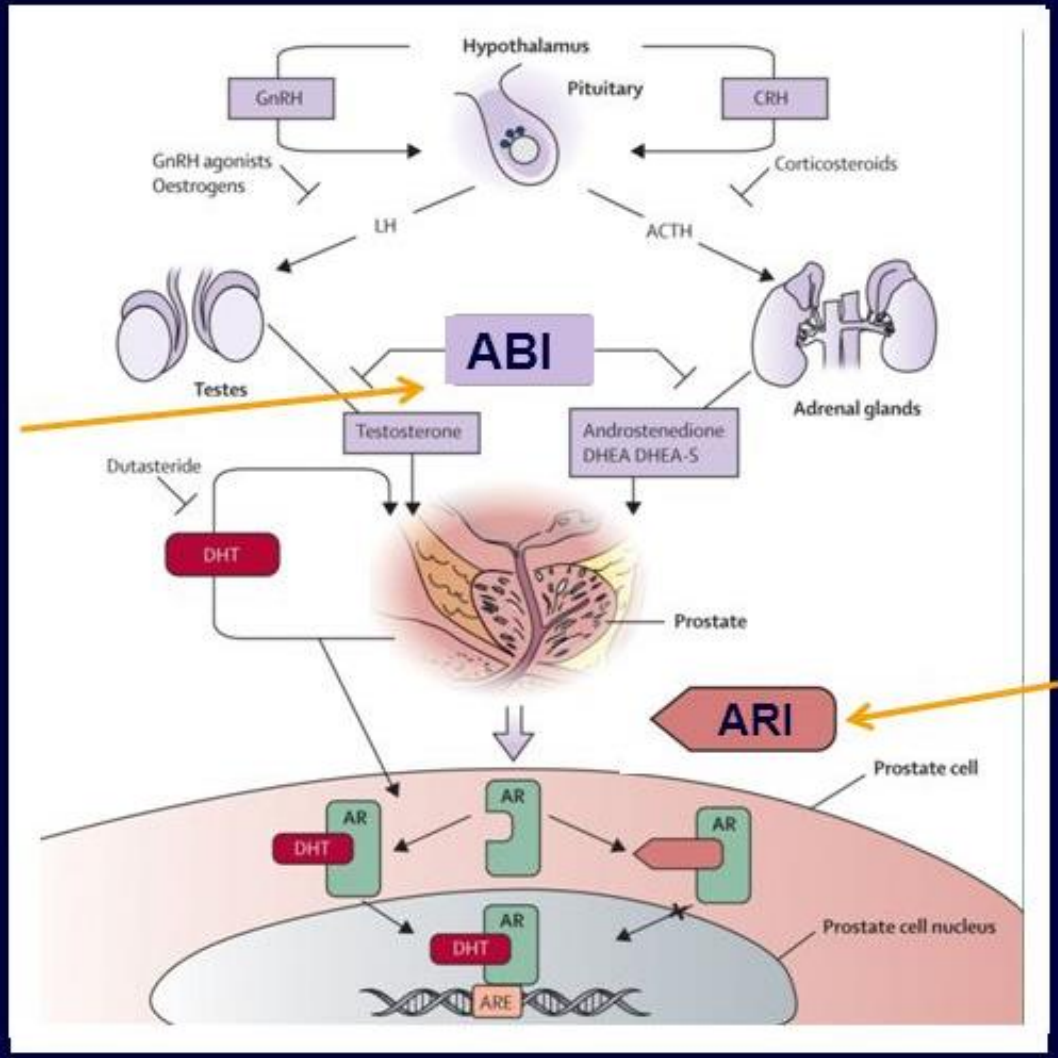
NAME	Half Life	Administration	AE/ monitoring
Flutamide (Eulexin ™)	6 hours	PO q8h	Induces cytochrome P450 and can cause hepatitis; **follow LFTs; gynecomastia;** GI upset and diarrhea
Bicalutamide (Casodex ™)	7 days	PO daily	**Follow LFTs; gynecomastia**
Nilutamide (Nilandron ™)	2 days	PO daily	**Follow LFTs; gynecomastia**; rare interstitial pneumonitis (follow CXR); visual disturbances in dark.

Second Generation Agents

Name	Class	Half life	Administration	Note
Abiraterone acetate (Zytiga™)	Androgen Biosynthesis Inhibitor (ABI)	12 hours	PO daily with prednisone PO BID (take with empty stomach)	Induces cytochrome P450 and can cause hepatitis; follow LFTs; caution with cardiac disease/CHF; can cause HTN, fluid retention and hypokalemia
Enzalutamide (Xtandi™)	2nd Gen Anti-androgen	6 days	PO daily	Induces cytochrome P450; contraindicated with seizures; association with ischemic heart disease and falls/fractures; rare PRES (posterior reversible encephalopathy); gynecomastia
Apalutamide (Erleada™)	2nd Gen Anti-androgen	3 days	PO daily	Induces cytochrome P450; Skin rash (25%); gynecomastia; contraindicated with seizures; association with falls/fractures and hypothyroidism
Darolutamide (Nubeqa™)	2nd Gen Anti-androgen	20 hrs	PO BID	Induces cytochrome P450; Doesn't cross blood/brain barrier – no contraindication with seizures or falls/fractures

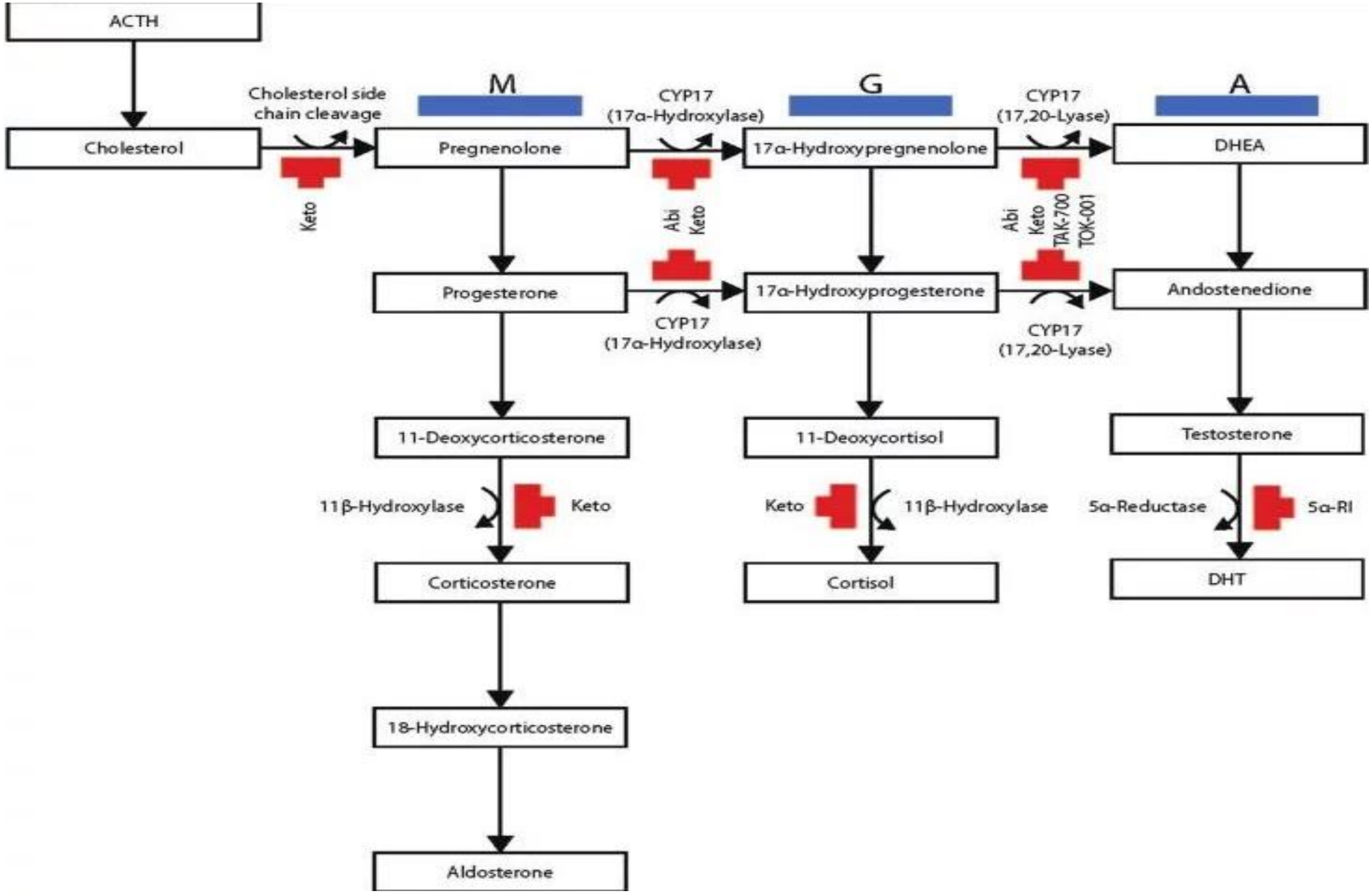
Androgen-Signaling Axis and Inhibitors

Androgen Biosynthesis Inhibitors (ABI):
Ketoconazole
Abiraterone
TAK700



Second generation AR inhibitors (ARI):
Enzalutamide (MDV3100)
ARN509

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ACTH
Cholesterol
Cholesterol side chain cleavage
Keto
M
Pregnenolone
CYP17 (17α-Hydroxylase)
Abi
Keto
G
17α-Hydroxypregnenolone
CYP17 (17,20-Lyase)
Abi
Keto
TAK-700
TOK-001
A
DHEA
Progesterone
CYP17 (17α-Hydroxylase)
17α-Hydroxyprogesterone
CYP17 (17,20-Lyase)
Andostenedione
11-Deoxycorticosterone
11β-Hydroxylase
Keto
Corticosterone
18-Hydroxycorticosterone
Aldosterone
11-Deoxycortisol
Keto
11β-Hydroxylase
Cortisol
Testosterone
5α-Reductase
5α-RI
DHT

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Can J Urol. 2010 Feb;17 Suppl 1:35-51. PMID: 20170599